

Individual Reflection - Lab1

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My tasks:

1. Search Microsoft Azure AI and collect the information about Machine Learning in Azure.
2. Find product and service about Machine Learning in Azure AI
3. Try some important product and service
4. Reading the documents about those product and service

Challenges:

In this group project, the greatest challenge was that we are novice in the machine learning field, both in theory and practice. There are 2 points about this challenge.

1. I don't really understand what collaboration is. Collaboration is one of four most important reasons why machine learning needs cloud computing. Unlike software development projects, the requirement is blueprint, developers write code to implement different parts of the blueprint. In machine learning, the 80/20 rule of data science is 80% data preparation, 20% model training. The codebase for machine learning is small and doesn't need team collaboration; one person can handle it alone. Data preparation may need multi-person collaboration. I can imagine that machine learning work in the cloud is an auto pipeline, but I still don't know how the collaboration happens until Gemini outputs the "AI Career Demand Heat Map". As a software developer, the roles and their core market shift give me an overview of what's collaboration in machine learning.

AI Career Demand Heat Map (2025-2026)

Role Name	Demand Heat	Core Market Shift	Salary Trend
Machine Learning Engineer (MLE)	🔥🔥🔥🔥🔥	From "Modeling" to "Systems." Focus is on performance tuning, quantization, and high-performance serving (Triton/vLLM).	High & Rising
AI / GenAI Engineer	🔥🔥🔥🔥🔥	The 2026 "Star Player." Specializes in RAG, Agentic Workflows, and integrating LLMs into existing software stacks.	Aggressive Growth
Data Engineer (DE)	🔥🔥🔥🔥	The "Clean Data" Pillar. Crucial for building Vector DBs and high-quality synthetic data pipelines for LLM training.	Steady & Critical
MLOps / AI Infra Engineer	🔥🔥🔥🔥	Scaling Excellence. Focus on K8s for AI, GPU orchestration, and CI/CD for automated model retraining.	Premium / Rare
Data Scientist (DS)	🔥🔥🔥	Specialization Over Generalization. Pivot toward Causal Inference, AI Safety, and Domain-Specific Research.	Bifurcated
AI Product Manager	🔥🔥🔥	The "Strategic Bridge." Defining LLM feasibility and ROI. Requires deep understanding of LLM limitations.	Rising

2. I am confused by the wide variety of machine learning products and services on Azure. As a machine learning beginner, I have explored several products and services on Azure. After reading documents, I only have a general understanding of how they work. However, I know there are **three pillars in Microsoft Azure AI**, and understanding these pillars is essential when studying or working with cloud computing. These three pillars also can apply to AWS, Google cloud platform as well as other popular AI cloud computing platforms.

The 3 Pillars of Microsoft Azure AI (2026 Edition)

Pillar	Focus & Core Capability	Typical Use Cases	Primary Azure Product/Service
Pillar 1: LLM (GenAI)	Reasoning & Planning. Focuses on understanding and generating human-like text, code, and complex reasoning.	Chatbots, RAG (Retrieval-Augmented Generation), automated coding, and AI Agents.	Azure AI Foundry (integrates Azure OpenAI, Llama, Mistral, and Claude).
Pillar 2: AI Services (APIs)	Universal Perception. "Out-of-the-box" sensory capabilities—seeing, hearing, and translating without needing to train a model.	OCR (Invoice/ID reading), Video safety monitoring, Real-time translation, Face recognition.	Azure AI Services (Formerly Cognitive Services): Vision, Speech, Language, and Document Intelligence.
Pillar 3: Custom ML (Classical AI)	Private Business Logic. Handling structured data, extreme long-tail scenarios, and high-frequency numerical forecasting.	Fraud detection, predictive maintenance (IoT), demand forecasting, and personalized recommendation engines.	Azure Machine Learning (AML), Azure Databricks, and AutoML.